C. Maximum Even Sum

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: brute force, greedy, implementation, math | https://codeforces.com/problemset/problem/2137/C

You are given two integers a and b. You are to perform the following procedure:

First, you choose an integer k such that b is divisible by k. Then, you simultaneously multiply a by k and divide b by k.

Find the greatest possible **even** value of $a+b$. If it is impossible to make $a+b$ even, output -1 instead.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers a and b ($1 \le a,b \le a \cdot b \le 10^{18}$).

Output

For each test case, output the maximum **even** value of $a+b$ on a new line.

Example

Input
7 8 1 1 8 7 7 2 6 9 16 1 6 4 6

Output
-1 6 50 8 74 -1 14

Note

In the first test case, it can be shown it is impossible for $a+b$ to be even.

In the second test case, the optimal k is 2. The sum is 2+4=6.